

Elementary & Intermediate Algebra



Solving linear equations

1 Solve for x : $4x - 9 = 23$.

2 Solve for x : $\frac{3x - 2}{5} = 8$.

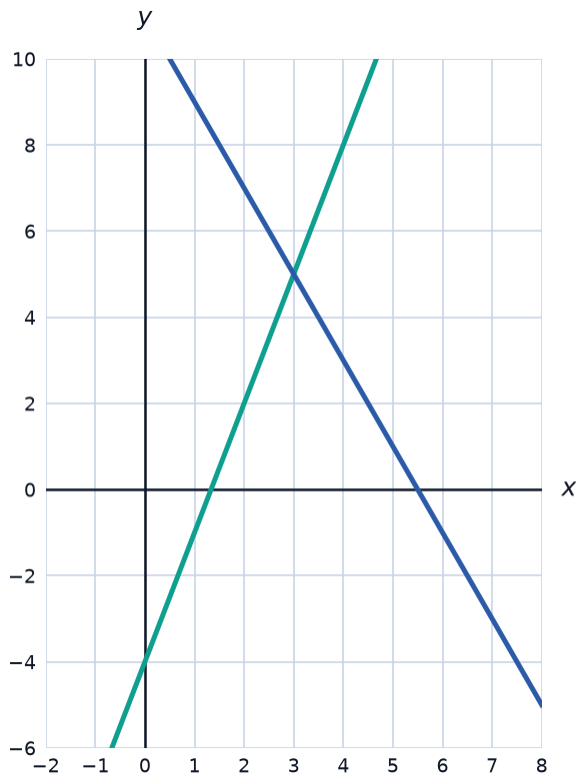
Solving and graphing inequalities

3 Solve $5x + 3 < 28$, giving your answer as an inequality.

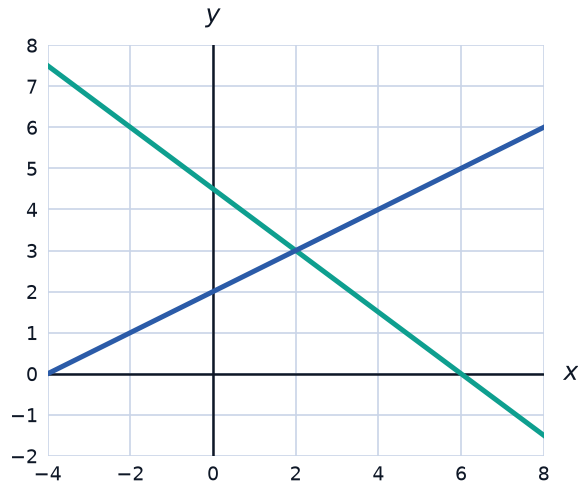
4 Solve $-4x + 7 \leq -9$, being careful with the direction of the inequality.

Systems of linear equations

5 Solve the system: $y = 3x - 4$ and $2x + y = 11$.



- 6 Solve the system: $3x + 4y = 18$ and $x - 2y = -4$, using elimination.



Factoring and solving quadratic equations

- 7 Solve $x^2 - x - 12 = 0$ by factoring.
- 8 Solve $2x^2 - 5x - 3 = 0$ by factoring.

Simplifying rational and radical expressions

- 9 Simplify $\frac{x^2 - 9}{x + 3}$.
- 10 Simplify $\sqrt{48}$.

Working with exponents

- 11 Simplify $\frac{x^7}{x^3}$.
- 12 Simplify $(3x^2y)^3$.

Absolute value equations

- 13 Solve $|3x + 2| = 11$.
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Word problems leading to linear or quadratic equations

- 14 The length of a rectangle is 3 more than twice its width, and its area is 65. Find the width and length.



- 15 A number increased by 8 is 3 times the original number decreased by 4. Find the number.
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Combining algebraic techniques in a multi-step problem

- 16 This question has two parts. Two consecutive positive integers have a product of 132. (a) Set up and solve a quadratic equation to find the two integers. (b) Verify your answer by checking the product.
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Solving quadratics using the quadratic formula

- 17 Solve $x^2 + 3x - 2 = 0$ using the quadratic formula, giving exact answers.
- 18 Use the discriminant to determine the number of real solutions of $4x^2 - 12x + 9 = 0$, without solving.
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Factoring by grouping and special patterns

- 19 Factor $x^3 + 2x^2 - 9x - 18$ by grouping.
- 20 Factor $4x^2 - 25$ completely.
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Rational equations and extraneous solutions

- 21 Solve $\frac{x}{x-2} = \frac{3}{x-2} + 1$, checking for extraneous solutions.
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Multi-step algebraic word problems

- 22 This question has two parts. A boat travels 60 miles downstream in 3 hours and returns upstream in 5 hours. (a) Let b be the boat's speed in still water and c be the current's speed; write two equations based on the trip times. (b) Solve the system to find b and c .
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Multiple choice — easy

- 23 Solve for x : $3x + 4 = 31$.

a. 10
b. 8
c. 9
d. 11

- 24 Solve for x : $9x + 8 = -46$.

a. -4
b. -7
c. -6
d. -5

- 25 Solve for x : $8x + 14 = 86$.

a. 9
b. 11
c. 10
d. 8

26 Solve for x : $4x + 14 = -14$.

a. -5

b. -7

c. -6

d. -8

27 Solve for x : $9x = 36$.

a. 4

b. 3

c. 5

d. 6

28 Solve for x : $2x + 8 = 26$.

a. 8

b. 10

c. 11

d. 9

29 Solve for x : $6x - 13 = 41$.

a. 8

b. 10

c. 9

d. 11

30 Solve for x : $4x - 13 = 15$.

a. 8

b. 6

c. 7

d. 9

31 Solve for x : $7x - 13 = 15$.

a. 6

b. 4

c. 5

d. 3

32 Solve for x : $5x - 14 = 16$.

a. 7

b. 8

c. 6

d. 5

33 Solve for x : $5x + 15 = -15$.

a. -4

b. -5

c. -7

d. -6

34 Solve for x : $9x + 17 = 125$.

a. 13

b. 12

c. 11

d. 14

35 Solve for x : $9x - 18 = 27$.

a. 7

b. 5

c. 4

d. 6

36 Solve for x : $6x = 48$.

a. 9

b. 7

c. 8

d. 10

37 Solve for x : $5x + 17 = -33$.

a. -11

b. -10

c. -8

d. -9

Multiple choice — medium

- 38 Solve $x^2 - 1 = 0$ by factoring. Give your answer as an ordered pair (smaller root first).
- a. $(1, -1)$
 - b. $(0, 1)$
 - c. $(-1, 1)$
 - d. $(-1, 0)$
- 39 Solve $x^2 - 49 = 0$ by factoring. Give your answer as an ordered pair (smaller root first).
- a. $(7, -7)$
 - b. $(-7, 6)$
 - c. $(-6, 7)$
 - d. $(-7, 7)$
- 40 Solve $x^2 + 2x - 8 = 0$ by factoring. Give your answer as an ordered pair (smaller root first).
- a. $(4, -2)$
 - b. $(-3, 2)$
 - c. $(-4, 1)$
 - d. $(-4, 2)$
- 41 Solve $x^2 - 6x - 16 = 0$ by factoring. Give your answer as an ordered pair (smaller root first).
- a. $(2, -8)$
 - b. $(-1, 8)$
 - c. $(-2, 7)$
 - d. $(-2, 8)$

- 42 Solve $x^2 - x - 42 = 0$ by factoring. Give your answer as an ordered pair (smaller root first).
- a. $(6, -7)$
 - b. $(-6, 7)$
 - c. $(-5, 7)$
 - d. $(-6, 6)$
- 43 Solve $x^2 + 4x - 60 = 0$ by factoring. Give your answer as an ordered pair (smaller root first).
- a. $(-10, 6)$
 - b. $(-9, 6)$
 - c. $(-10, 5)$
 - d. $(10, -6)$
- 44 Solve $x^2 - 7x - 8 = 0$ by factoring. Give your answer as an ordered pair (smaller root first).
- a. $(0, 8)$
 - b. $(-1, 7)$
 - c. $(1, -8)$
 - d. $(-1, 8)$
- 45 Solve $x^2 + 17x + 70 = 0$ by factoring. Give your answer as an ordered pair (smaller root first).
- a. $(-10, -7)$
 - b. $(-9, -7)$
 - c. $(-10, -8)$
 - d. $(10, 7)$

46 Solve $x^2 + 4x - 21 = 0$ by factoring. Give your answer as an ordered pair (smaller root first).

a. $(-6, 3)$

b. $(-7, 3)$

c. $(7, -3)$

d. $(-7, 2)$

47 Solve $x^2 - 4x - 5 = 0$ by factoring. Give your answer as an ordered pair (smaller root first).

a. $(1, -5)$

b. $(-1, 5)$

c. $(-1, 4)$

d. $(0, 5)$

48 Solve $x^2 + x - 2 = 0$ by factoring. Give your answer as an ordered pair (smaller root first).

a. $(-1, 1)$

b. $(-2, 0)$

c. $(-2, 1)$

d. $(2, -1)$

49 Solve $x^2 - 11x + 18 = 0$ by factoring. Give your answer as an ordered pair (smaller root first).

a. $(2, 9)$

b. $(3, 9)$

c. $(2, 8)$

d. $(-2, -9)$

50 Solve $x^2 - 16x + 63 = 0$ by factoring. Give your answer as an ordered pair (smaller root first).

a. (7, 8)

b. (-7, -9)

c. (7, 9)

d. (8, 9)

51 Solve $x^2 - 3x = 0$ by factoring. Give your answer as an ordered pair (smaller root first).

a. (1, 3)

b. (0, 2)

c. (0, -3)

d. (0, 3)

52 Solve $x^2 - 25 = 0$ by factoring. Give your answer as an ordered pair (smaller root first).

a. (-5, 4)

b. (5, -5)

c. (-4, 5)

d. (-5, 5)

Multiple choice — hard

53 Solve $x^2 + 8x - 8 = 0$ using the quadratic formula, giving exact answers.

a. $-4 \pm 2\sqrt{6}$

b. $-3 \pm 2\sqrt{6}$

c. $-4 \pm 3\sqrt{6}$

d. $4 \pm 2\sqrt{6}$

54 Solve $x^2 + 14x - 1 = 0$ using the quadratic formula, giving exact answers.

a. $-6 \pm 5\sqrt{2}$

b. $-7 \pm 5\sqrt{2}$

c. $7 \pm 5\sqrt{2}$

d. $-7 \pm 6\sqrt{2}$

55 Solve $x^2 - 10x - 9 = 0$ using the quadratic formula, giving exact answers.

a. $6 \pm \sqrt{34}$

b. $5 \pm 2\sqrt{34}$

c. $5 \pm \sqrt{34}$

d. $-5 \pm \sqrt{34}$

56 Solve $x^2 - 10x + 4 = 0$ using the quadratic formula, giving exact answers.

a. $6 \pm \sqrt{21}$

b. $5 \pm \sqrt{21}$

c. $5 \pm 2\sqrt{21}$

d. $-5 \pm \sqrt{21}$

57 Solve $x^2 + 2x - 2 = 0$ using the quadratic formula, giving exact answers.

a. $1 \pm \sqrt{3}$

b. $-1 \pm 2\sqrt{3}$

c. $-1 \pm \sqrt{3}$

d. $0 \pm \sqrt{3}$

58 Simplify $(x^2)^3$.

a. x^8

b. x^6

c. x^5

d. x^7

59 Simplify x^7/x^3 .

a. x^5

b. x^3

c. x^4

d. x^6

60 Simplify x^{10}/x^9 .

a. x^2

b. x^3

c. x^1

d. x^4

61 Simplify x^9/x^7 .

a. x^3

b. x^4

c. x^2

d. x^1

62 Simplify $(x^4)^3$.

a. x^{12}

b. x^{14}

c. x^{11}

d. x^{13}

63 Solve the system: $3x + 4y = 26$ and $5x + y = 15$. Find the value of x .

a. 1

b. 2

c. 5

d. 3

64 Solve the system: $x + 6y = -41$ and $2x + y = -16$. Find the value of x .

a. -3

b. -4

c. -6

d. -5

65 Solve the system: $2x + 3y = 2$ and $4x + y = 14$. Find the value of x .

a. -2

b. 5

c. 4

d. 3

66 Solve the system: $3x + 6y = 3$ and $6x + 3y = -12$. Find the value of x .

a. -2

b. -3

c. -4

d. 2

67 Solve the system: $2x + 4y = 18$ and $6x + 4y = 22$. Find the value of x .

a. 2

b. 4

c. 0

d. 1